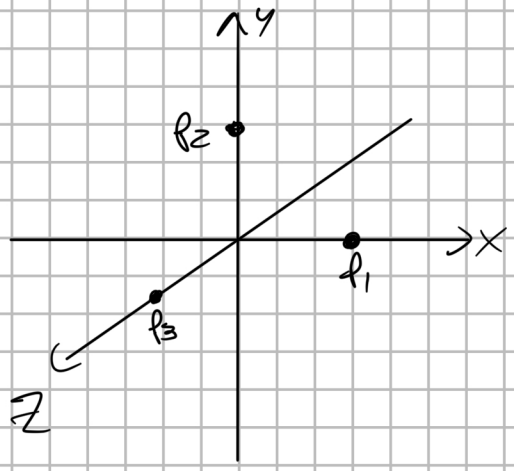
7- $\vec{dl} = 2\hat{k}\ mm$ (eje z)

$i = 2A$

$P_1 = (3,0,0)\ m$

$P_2 = (0,3,0)\ m$

$P_3 = (0,0,3)\ m$



$$\vec{dB} = \frac{\mu_0\, i\, \vec{dl}}{4\pi r^2} \times \hat{r} \qquad r_1^2 = r_2^2 = r_3^2 = 9m^2$$

$\vec{i} \times \vec{j} = \vec{k}$

$\vec{j} \times \vec{k} = \vec{i}$

$\vec{k} \times \vec{i} = \vec{j}$

$\mu_0 = 4\pi \cdot 10^{-7}\ H/m$

a- $$\vec{dB_1} = \frac{\mu_0\, i \cdot 0{,}002m \cdot \vec{k} \times \vec{i}}{36\pi m^2}$$

$$\vec{dB_1} = \frac{\mu_0 \cdot 0{,}004m \cdot A \cdot \vec{j}}{36\pi m^2} = \frac{4}{9} \cdot 10^{-10}\, \vec{j}\ T$$

b- $$\vec{dB_2} = \frac{\mu_0 \cdot i \cdot 0{,}002m \cdot \vec{k} \times \vec{j}}{36\pi m^2}$$

$$\vec{dB_2} = \frac{\mu_0 \cdot 0{,}004\, mA}{36\pi m^2} \cdot \vec{i} = -\frac{4}{9} \cdot 10^{-10}\, \vec{i}\ T$$

c- $$\vec{dB_3} = \frac{\mu_0 \cdot i \cdot 0{,}002m \cdot \vec{k} \times \vec{k}}{36\pi m^2} = \vec{0}\ T$$

d- $\vec{dl_2} = 1\hat{i}\ mm = 0{,}001\hat{i}\ m$ $\quad \vec{dl_1} = 2\hat{k}\ mm = 0{,}002\hat{k}\ m$

$i_2 = 3A$ $\quad i_1 = 2A$

$$\vec{dF_{12}} = I_2 \vec{dl_2} \times \vec{dB_1} \qquad \vec{dB_1} = \frac{4}{a} \cdot 10^{-10}\hat{j}\ T$$

$$\vec{dF_{12}} = \frac{4}{3} \cdot 10^{-13}\hat{k}\ N$$

e - $\vec{dF_{12}} = I_1 \vec{dl_1} \times \vec{dB_2}$ $\quad \vec{r} = -3\hat{i}$

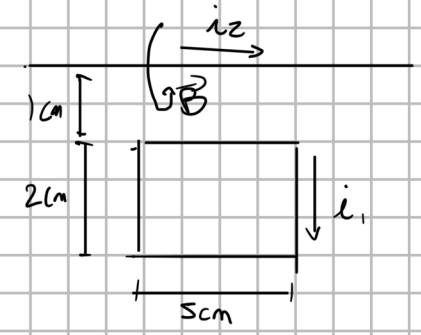
$$\vec{dB_2} = \frac{\mu_0 \cdot I_2 \vec{dl_2} \times \check{r}}{4\pi r^2}$$

$$\vec{dB_2} = \frac{-\mu_0 \cdot I_2 \cdot 0{,}001\,m\ \hat{i} \times \hat{i}}{4\pi r^2} = \vec{0}\ T$$

$$\vec{dF_{21}} = I_1 \vec{dl_1} \times \vec{dB_2}$$

$$= 2(2 \cdot 10^{-3}\hat{k})m \times \vec{0}\ T = 0N$$

2 -

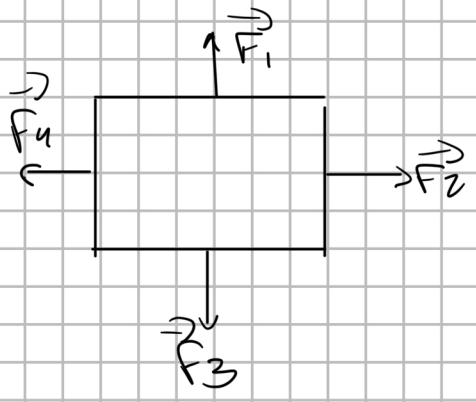


$i_1 = i_2 = 10A$

$$|\vec{B}| = \frac{\mu_0 I_1}{2\pi r} \qquad \vec{B} = \frac{-\mu_0 I_1}{2\pi r}\hat{k}$$

$$\vec{F} = I_2 \vec{L} \times \vec{B}$$

$$B_1 = \frac{\mu_0 \cdot 10A}{2\pi \cdot 0{,}01m} = 2 \cdot 10^{-4} T$$



$$B_3 = \frac{\mu_0 \cdot 10A}{2\pi \cdot 0{,}03m} = 6{,}67 \cdot 10^{-5} T$$

$$F_1 = I_2 L B_1 = 10 \cdot 0{,}05 \cdot 2 \cdot 10^{-4} N = 10^{-4} N$$

$F_2 = -F_4$
(se cancelan)

$$F_3 = I_2 L B_3 = 10 \cdot 0{,}05 \cdot 6{,}67 \cdot 10^{-5} T = 3{,}33 \cdot 10^{-5} N$$

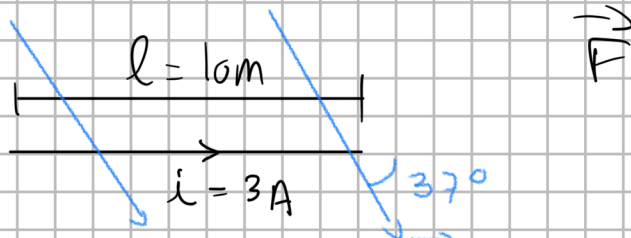
$$\vec{F} = \vec{F_1} + \cancel{\vec{F_2}} + \vec{F_3} + \cancel{\vec{F_4}}$$

$$= 10^{-4} \cdot \hat{j} N - 3{,}33 \cdot 10^{-5} \hat{j} N$$

$$= 6{,}67 \cdot 10^{-5} \hat{j} N$$

apunta hacia arriba (atracción)

3 -

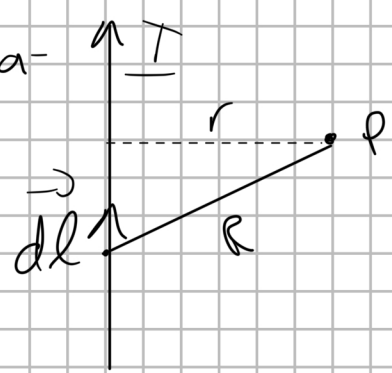


$$\vec{F} = I\vec{L} \times \vec{B}$$

$$= I L B \operatorname{Sen}(37°)$$

$$= 27 N$$

4.a-

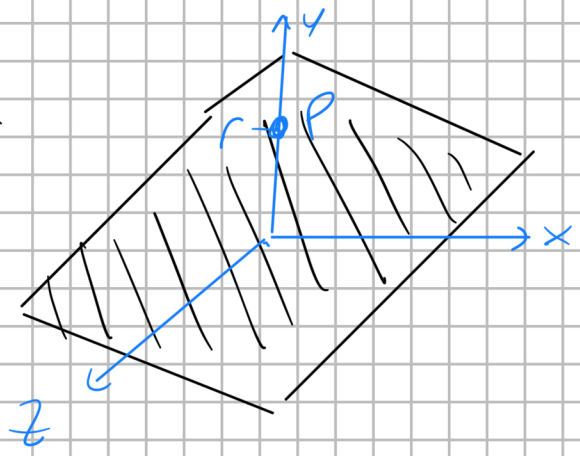


$$d\vec{B} = \frac{\mu_0 I \vec{dl} \times \hat{r}}{4\pi r^2}$$

$$B = \frac{\mu_0 I}{4\pi} \int_{-\infty}^{+\infty} \frac{r\, dz}{(r^2+z^2)^{3/2}} = \frac{\mu_0 I}{2\pi r}$$

dirección B = Tangencial a circunferencias concéntricas alrededor del hilo

b-



plano XZ cargado con corriente K [A/m]

El plano es un conj de lineas infinitas

$$dI = K\,dx$$

$$dB = \frac{\mu_0 dI}{2\pi R} \Rightarrow dB = \frac{\mu_0\, dI}{2\pi\sqrt{x^2+r^2}} \qquad \cos\theta = \frac{r}{R}$$

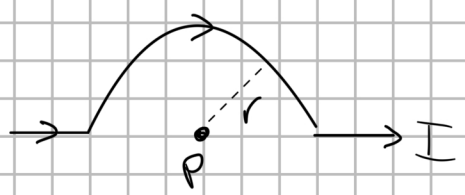
$$R = \sqrt{x^2+r^2}$$

$$dB_x = dB\cos\theta = \frac{\mu_0 K r}{2\pi(x^2+r^2)}\,dx$$

$$B_x = \frac{\mu_0 K r}{2\pi}\int_{-\infty}^{\infty}\frac{dx}{r^2+x^2} = \frac{\mu_0 K r}{2\pi}\left[\frac{1}{r}\left(\frac{\pi}{2}-\left(-\frac{\pi}{2}\right)\right)\right]$$

$$B_x = \frac{\mu_0 K}{2\pi}\cdot\pi = \frac{\mu_0 K}{2}$$

4-



$I = 15A \qquad r = 20cm$

$$\vec{B} = \vec{B}_{izq} + \vec{B}_{semi} + \vec{B}_{der}$$

$$\vec{B} = \frac{\mu_0 I \, d\vec{l} \times \vec{r}}{4\pi r^2}$$

$$d\vec{l} \times \vec{r} = dl \cdot r \cdot \operatorname{sen}\theta$$

$\theta_{izq} = 180° \Rightarrow B_{izq} = 0$

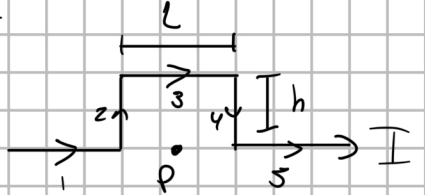
$\theta_{der} = 0° \Rightarrow B_{der} = 0$

$\theta_{semi} = 90°$

$$B = \frac{\mu_0 I}{4\pi r^2} \int dl$$

$$B = \frac{\mu_0 I}{4\pi r^2} \cdot \pi r = \frac{\mu_0 I}{4r} = 2{,}36 \cdot 10^{-5} \, T$$

6 -



$I = 8A \quad L = 2cm \quad h = 1cm$

$\vec{B} = \vec{B_1} + \vec{B_2} + \vec{B_3} + \vec{B_4} + \vec{B_5}$

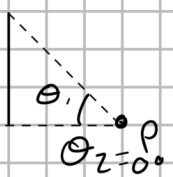
$\vec{B_1} = \vec{B_5} = \vec{0} \, T$

$$B = \frac{\mu_0 I}{4\pi a} (\operatorname{sen}\theta_1 + \operatorname{sen}\theta_2) \longrightarrow$$

a: dist perpendicular desde P al conductor

θ_1 y θ_2: ángulos desde esa perpendicularidad hasta los extremos del tramo

$a_2 = a_4 = 0{,}01 m$



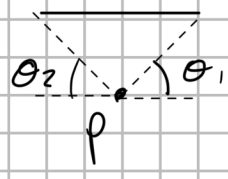
$\theta_1 = 45° \Rightarrow B_2 = \frac{\mu_0 I}{4\pi a} (\operatorname{sen} 45° + \operatorname{sen} 0°)$

$B_2 = B_4 = 5{,}66 \cdot 10^{-5} \, T$

(ambos entrantes a la hoja por regla de la mano der.)

$a_3 = 0{,}01 m$

$\theta_1 = \theta_2 = 45°$



$B_3 = \frac{\mu_0 I}{2\pi a} \operatorname{sen} 45° = 1{,}13 \cdot 10^{-4} \, T$ (También entrante)

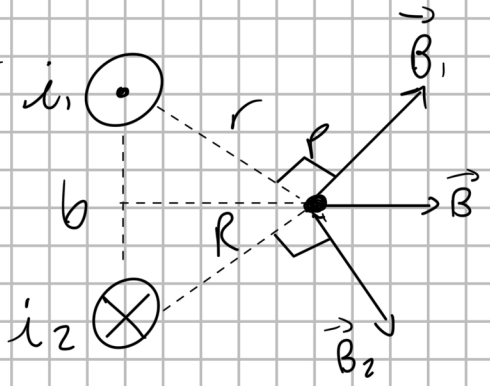
$$\vec{B} = \cancel{\vec{B}_1} + \vec{B}_2 + \vec{B}_3 + \vec{B}_4 + \cancel{\vec{B}_5}$$

$$= -(2 \cdot 5{,}66 \cdot 10^{-5} + 1{,}13 \cdot 10^{-4})\,\hat{k}\ T$$

$$= -2{,}26 \cdot 10^{-4}\,\hat{k}\ T$$

7- $\vec{F} = I(\vec{L} \times \vec{B})$

La fuerza $\vec{F}$ magnética depende de $i = \frac{dq}{dt}$, no de q

8- 

$$r = \sqrt{R^2 + \frac{b^2}{4}} \qquad i_1 = i_2 = i$$

$$B_1 = B_2 = \frac{\mu_0 i}{2\pi\sqrt{R^2 + b^2/4}}$$

$\vec{B} = \vec{B}_1 + \vec{B}_2$

$$B = \frac{\mu_0 i}{\pi\sqrt{R^2+b^2/4}} \cdot \cos\theta = \frac{\mu_0 i}{\pi\sqrt{R^2+b^2/4}} \cdot \frac{b/2}{r} = \frac{\mu_0 i\, b}{2\pi(R^2+b^2/4)}$$

9.a- $\vec{F} = q(\vec{v} \times \vec{B}) \Rightarrow F = |q|\, vB \operatorname{sen}\theta$

manteniendo $vB\operatorname{sen}\theta$ cte $\Rightarrow F \propto |q|$. Verdadero

b- Falso.

$\vec{v} \times \vec{B}$ nos da $\vec{F}$ perpendicular a ambos

$\Rightarrow \vec{F} \perp \vec{v}$

10.a- Falso. $1\ gauss = 10^{-4}\ T$

b- verdadero.

$\vec{F} = I(\vec{L} \times \vec{B}) \Rightarrow \vec{F} \perp \vec{L}$

Como la fuerza es perpendicular a la trayectoria, F no Trabaja.

11- $B_x = 0\ T$ $\quad \vec{v} = (40\hat{i} + 35\hat{j})\ km/s = (4 \cdot 10^4 \hat{i} + 3{,}5 \cdot 10^4 \hat{j})\ m/s$

$\vec{F} = (-4{,}2 \cdot 10^{-12}\hat{i} + 4{,}8 \cdot 10^{-12}\hat{j})\ N$

$q = q_{e^-} = -1{,}6 \cdot 10^{-19}\ C$

$\vec{F} = q(\vec{v} \times \vec{B})$

$F_x = q(v_y B_z - \cancel{v_z B_y}^{\,0}) \Rightarrow B_z = \dfrac{F_x}{q v_y} = 750\ T$

$\cancel{F_z}^{\,0} = q(v_x B_y - \cancel{v_y B_x}^{\,0}) \Rightarrow B_y = 0\ T$

$\vec{B} = \vec{B_x} + \vec{B_y} + \vec{B_z} = 750\ \hat{k}\ T$

12- $q = q_p = 1{,}6 \cdot 10^{-19}\ C$

$m = m_p = 1{,}67 \cdot 10^{-27}\ kg$

$\vec{v} = 4 \cdot 10^6\ \hat{i}\ m/s$

$\vec{B} = 2\hat{k}\ T$

$\vec{F} = q(\vec{v} \times \vec{B})$

$\vec{F} = qvB(\hat{i} \times \hat{k})$

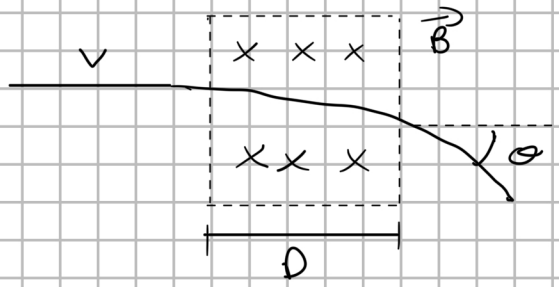
$\vec{F} = -qvB\hat{j}$

$\vec{F} = -1{,}28 \cdot 10^{-12}\hat{j}\ N$

$F_m = F_c$

$qvB = \dfrac{mv^2}{r} \Rightarrow r = \dfrac{mv}{qB} = 2{,}09\ cm$

13 -



$q = e$

$\Delta U = q \Delta V$

Se transforma en cinética

$eV = \frac{1}{2} m v^2$

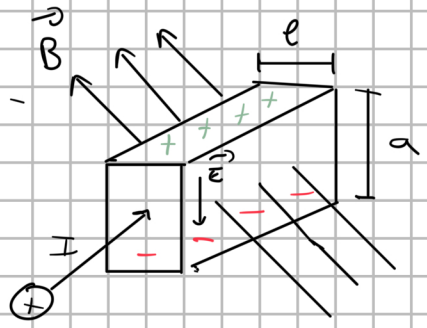
$F = evB = \frac{mv^2}{r}$ (centrípeta)

$v = \sqrt{\frac{2eV}{m}}$

$r = \frac{mv}{eB} = \frac{m}{eB}\sqrt{\frac{2eV}{m}} = \frac{\sqrt{2evm}}{eB}$

$\text{sen}\,\theta = \frac{D}{r} = \frac{DeB}{\sqrt{2emV}} \propto \frac{1}{\sqrt{V}}$

14 -



e: espesor

$\vec{F_m} = q(\vec{v} \times \vec{B})$

$\vec{F_e} = q\vec{E}$

En equilibrio:

$q(\vec{v} \times \vec{B}) = -q\vec{E}$

$\vec{v} \times \vec{B} = -\vec{E}$

$E = vB$

$\Delta V = -\int_0^a \vec{E}\, d\vec{\ell} = Ea$

$\Delta V = vBa$

$I = nqvA_\perp$

$A = ae$

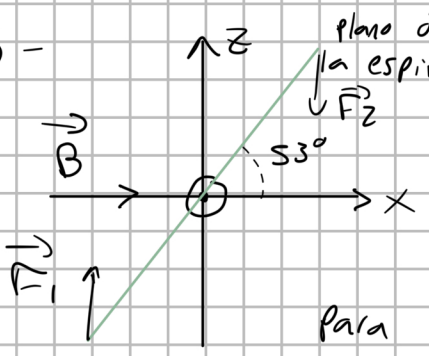
$\Rightarrow I = nqvae$

$v = \frac{I}{nqae}$

$\Delta V = \frac{IBa}{nqae}$

$\Delta V = \frac{IB}{nqe}$

15 -



$B = 0,2T \quad L = 0,05m$

$I = SA$

$\vec{F} = I(\vec{L} \times \vec{B})$

Para los laterales de la espira (paralelos al eje y)

$\hat{j} \perp \hat{i} \Rightarrow F_1 = F_2 = ILB$

Para los otros lados:

$F_3 = F_4 = ILB \operatorname{sen}(53°)$

Las fuerzas F_3 y F_4 son paralelas al eje y $\therefore$ no generan τ

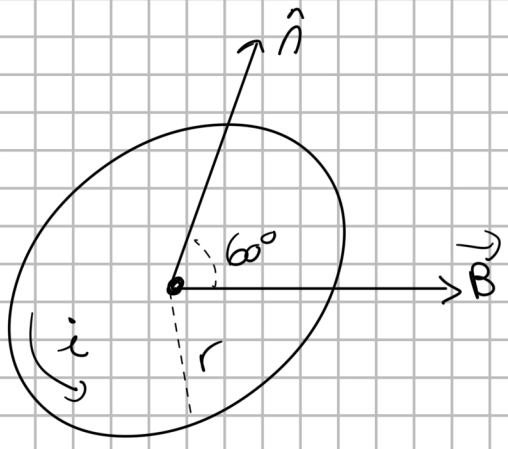
$d = \frac{L}{2}\cos 53° \qquad \tau = F \cdot d$

$\tau_1 = F_1 \cdot d_1 = ILB\,d$

$\tau_2 = F_2 d_2 = ILBd$

$\tau = \tau_1 + \tau_2 = 2ILBd \simeq IBL^2 \cos 53°$

16 -



$N = 20$ vueltas $\quad r = 0,04m$

$i = 3A \qquad B = 5000$ gauss

$B = 0,5T$

$\vec{m} = I\vec{A}$

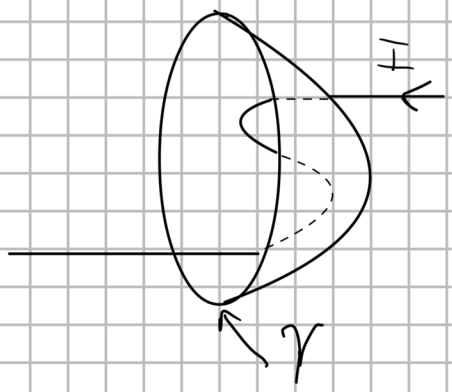
$m = NIA = NI\pi r^2 = 0,3\,Am^2$

$\vec{\tau} = \vec{m} \times \vec{B} = mB \operatorname{sen} 60° = 0,13\,N \cdot m$

17 - Ley de Ampere: $\oint_\gamma \vec{B} \cdot \vec{dl} = \mu_0 i_{enc}$

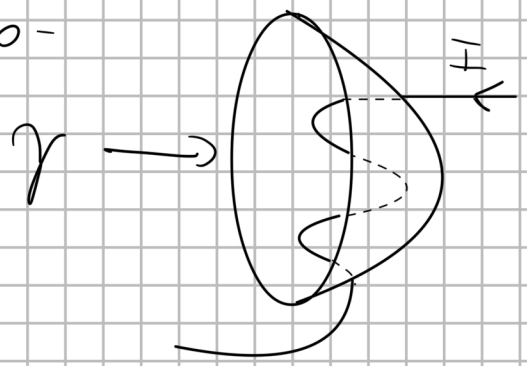
Falso. No depende de la forma que tenga la corriente

18. a-



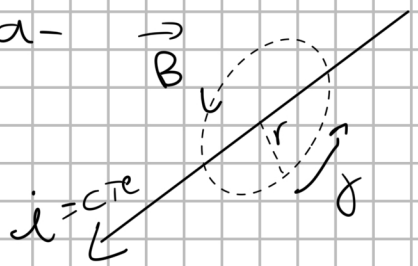
$$\oint_\gamma \vec{B}\cdot d\vec{\ell} = \mu_0 i_{enc}$$
$$= \mu_0 (i - i + i)$$
$$= \mu_0 i$$

b-



$$\oint_\gamma \vec{B}\cdot d\vec{\ell} = \mu_0 i_{enc}$$
$$= \mu_0 (i - i + i - i)$$
$$= 0$$

19. a-

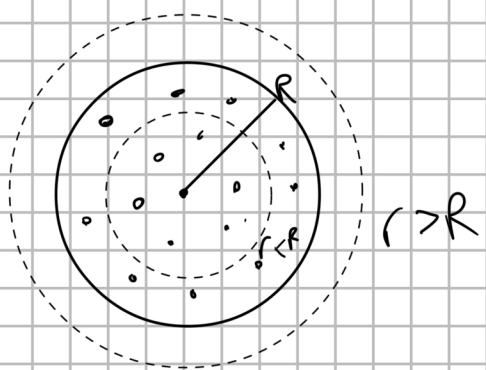


$$\oint_\gamma \vec{B}\cdot d\vec{\ell} = B \cdot \int_\gamma d\ell = B \cdot 2\pi r$$

$$B \cdot 2\pi r = \mu_0 i$$

$$B = \frac{\mu_0 i}{2\pi r}$$

b-



Para $r > R$:

$$\oint_\gamma \vec{B}\cdot d\vec{\ell} = \mu_0 I$$

$$B = \frac{\mu_0 I}{2\pi r}$$

Para $r < R$:

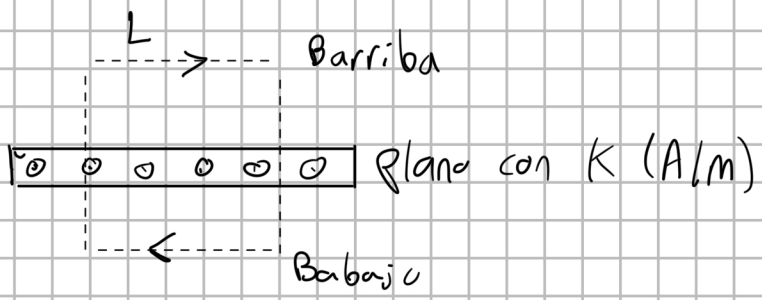
$$J = \frac{I}{\pi R^2}$$

$$\oint_\gamma \vec{B}\cdot d\vec{\ell} = \mu_0 I_{enc}$$

$$B(2\pi r) = \mu_0 J \cdot \pi r^2 = \mu_0 I \frac{r^2}{R^2}$$

$$B = \frac{\mu_0 I r}{2\pi R^2}$$

c-

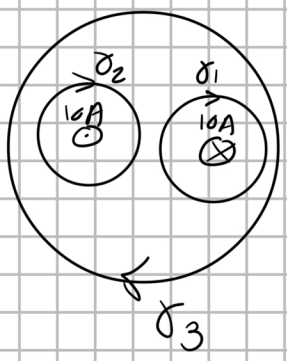


$$\oint_\gamma \vec{B}\cdot d\vec{\ell} = BL + BL = 2BL = \mu_0 I_{enc}$$

$$2BL = \mu_0 KL$$

$$B = \frac{\mu_0 K}{2}$$

20-



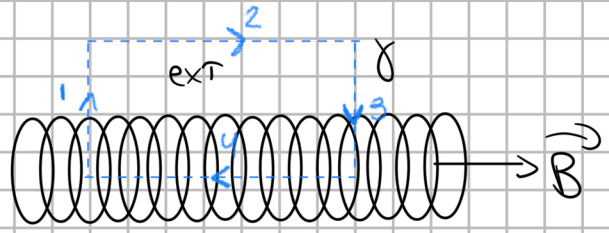
a- $\oint_{\gamma_1} \vec{B}\cdot d\vec{\ell} = \mu_0 I = 4\pi\cdot 10^{-6}\ T\cdot m$

$\oint_{\gamma_2} \vec{B}\cdot d\vec{\ell} = \mu_0(-I) = -4\pi\cdot 10^{-6}\ Tm$

$\oint_{\gamma_3} \vec{B}\cdot d\vec{\ell} = \mu_0(I-I) = 0$

b- $\vec{B}_T = \vec{B}_1 + \vec{B}_2$

21-



$$\oint_\gamma \vec{B}\cdot d\vec{\ell} = \mu_0 I$$

$$\int_1 \vec{B}\cdot d\vec{\ell} + \int_2 \vec{B}\, d\vec{\ell} + \int_3 \vec{B}\cdot d\vec{\ell} + \int_4 \vec{B}\cdot d\vec{\ell}$$

En el ext: $\vec{B} = 0 \Rightarrow \int_2 \vec{B}\cdot d\vec{\ell} = 0$

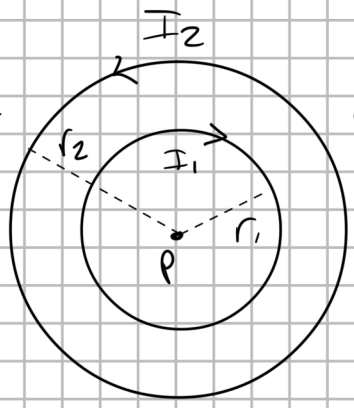
$$\int_4 \vec{B}\cdot d\vec{\ell} = \int B\cdot d\ell \cdot \operatorname{Sen} 0° = 0$$

$n = \frac{N}{L}$ (espiras por unidad de long)

$$\oint_\gamma \vec{B}\cdot d\vec{\ell} = B\ell = \mu_0 I_{enc} = \mu_0 N_{inc} I = \mu_0 n \ell I$$

$$B = \mu_0 n I = \mu_0 I \frac{N}{L}$$

22 -



a - $I_1 = 5A \quad R_1 = 0,1m \quad R_2 = 0,15m$

$$B = \frac{\mu_0 I}{2R} \Rightarrow B_1 = \frac{\mu_0 I_1}{2R_1} \; ; \; B_2 = \frac{\mu_0 I_2}{2R_2}$$

$$\vec{B} = \vec{B}_1 + \vec{B}_2 = 0 \Rightarrow B_1 = B_2 \Rightarrow \frac{\mu_0 I_1}{2R_1} = \frac{\mu_0 I_2}{2R_2}$$

$$\vec{B}_1 = -\vec{B}_2$$

$$-B_1 \cdot \hat{k} = -B_2 \hat{k}$$

$$I_2 = \frac{R_2}{R_1} I_1 = 7,5A$$

b - $\vec{m} = I\vec{A} \Rightarrow m = I \pi R^2$

$m_1 = I_1 \pi R_1^2 = 0,157 Am^2$ (entrante)

$m_2 = I_2 \pi R_2^2 = 0,53 Am^2$ (Saliente)

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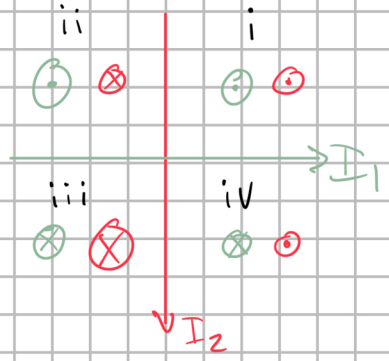

a- Se podrían cancelar los campos en los cuadrantes ii y iv

$$B = \frac{\mu_0 i}{2\pi r} \qquad B_1 = \frac{\mu_0 i}{2\pi |y|} \qquad B_2 = \frac{\mu_0 i}{2\pi |x|}$$

$$B_1 = B_2 \Rightarrow \frac{\mu_0 i}{2\pi |y|} = \frac{\mu_0 i}{2\pi |x|} \Rightarrow \frac{1}{|y|} = \frac{1}{|x|}$$

$i_1 = i_2 = i$

$$|y| = |x|$$

Como estamos en cuadrantes ii y iv $\Rightarrow y = -x$